COURSE 1

Quasicrystals, Diophantine approximation and algebraic numbers

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Abstract. Quasicrystals can be characterized by a remarkable Diophantine approximation property. This permits to define the dual quasicrystal Λ^* as the collection of y in $\mathbb{R}^n$ such that $|e^{iy\cdot x}-1| \leq 1$ for each x in the given quasicrystal Λ. In many cases one obtains $\Lambda^{**} = \Lambda$ and this duality is nicely related to the spectral properties of quasicrystals.

1. Introduction.

In the late sixties, I was studying some problems in harmonic analysis in which Pisot numbers were playing a key role. A crucial issue was to relate the arithmetical structure of a given set Λ of real numbers to the properties of the vector space consisting of all trigonometric sums $f(x) = \sum_{\lambda\in\Lambda} c_\lambda e^{i\lambda x}$ whose frequencies λ belong to Λ. The model case is the trivial observation that $f(x)$ is 2π-periodic when $\Lambda = \mathbf{Z}$.

This was the problem. Then I discovered some remarkable patterns Λ for

F. Axel et al. (eds.), *Beyond Quasicrystals*
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which all the above mentioned trigonometric sums satisfy

$$\sup_x |f(x-\tau)-f(x)| \leq \varepsilon \sup_x |f(x)| \tag{1}$$

for each τ belonging to a Delaunay set Λ_ε^*. Here ε is an arbitrary positive number and the *Delaunay set* Λ_ε^* does not depend on f. (The definition of a Delaunay set is given at the next page and (1) is detailed in section 4). Using a clever lemma by N. Varopoulos, I proved the equivalence between (1) and a *Diophantine approximation* property satisfied by Λ. I could characterize these sets Λ by a deceivingly simple additive property (definition 2 below). I wrote two books on the subject and gave an invited lecture at the ICM70 (Nice) on these patterns Λ which I named "model sets".

Fifteen years later on, two physicists working at Ecole Polytechnique, Michel Duneau and André Katz told me that my "model sets" anticipated Penrose tiling and quasicrystals.

Françoise Axel convinced me to lecture on my book during this conference on quasicrystals. Indeed most of the content of my book was still unknown to the participants.

Let me use this opportunity to offer the reader a leasury tour and let him have distinct viewpoints on quasicrystals. If needed, the proofs can be found in [**1**] or [**2**]. These notes will be divided into four parts. In the first one, some basic facts on almost periodic functions, Poisson summation formula and Pisot or Salem numbers will be stated, for the reader's convenience.

In the second section, the following definition will be used.

Definition 1. *A subset Λ of $\mathbb{R}^n$ is a Delaunay set if there exist two radii $R_2 > R_1 > 0$ such that each ball with radius R_1, whatever be its location, shall contain at most one point in Λ while each ball with radius R_2, whatever be its location, shall contain at least one point in Λ.*

The first requirement can be given the following equivalent formulation : there exists a positive r such that $\lambda \in \Lambda$, $\lambda' \in \Lambda$ and $\lambda \neq \lambda'$ imply $|\lambda'-\lambda| \geq r$.

A collection of points fulfilling the second condition is relatively dense (this terminology was introduced by Besicovitch).

Returning to quasicrystals, they will be defined by several equivalent properties. The first definition stresses the connection between a quasicrystal and a lattice. A lattice Λ contained in $\mathbb{R}^n$ is an additive subgroup. This can be written $\Lambda-\Lambda \subset \Lambda$ where $\Lambda-\Lambda$ is the collection of all $\lambda-\lambda'$, $\lambda \in \Lambda$, $\lambda' \in \Lambda$ and $\subset$ denotes inclusion. But a lattice Λ is also a Delaunay set. We are ready for the first definition.

Definition 2. *A quasicrystal Λ is a Delaunay set in $\mathbb{R}^n$ such that $\Lambda - \Lambda \subset \Lambda + F$ where F is a finite set.*

This approach is great since it is elegant and concise. But for everyday life one needs a more handable definition. This second definition is the celebrated "cut and projection" definition which was proved in [**2**] to be equivalent to definition 2, modulo finite sets.

In the third section, the same quasicrystals will be characterized by the same Delaunay property and by a remarkable *Diophantine approximation* property. This approach leads to the notion of the dual quasicrystal. The *dual lattice* Λ^* is the collection of all $y \in \mathbb{R}^n$ such that $\exp(iy \cdot \lambda) = 1$ for every λ in the given lattice Λ. In the case of arbitrary quasicrystal this definition will force y to be 0. Indeed, the *dual quasicrystal* Γ^* is defined as

$$\Lambda^* = \left\{ y \in \mathbb{R}^n \; ; \; |\exp(iy \cdot \lambda) - 1| \leq 1 \, , \; \lambda \in \Lambda \right\} .$$

In many cases, we will obtain $(\Lambda^*)^* = \Lambda$ as expected.

In the last section, a *Poisson summation formula* will be given in the quasicrystal setting. Roughly speaking the Fourier transform of a sum of Dirac masses δ_λ, $\lambda \in \Lambda$, is a sum of Dirac masses δ_y where y runs over the dual quasicrystal Λ^*. This statement becomes a true fact if the Dirac masses δ_λ, $\lambda \in \Lambda$, are suitably weighted. We then consider the measure $\mu = \sum_{\lambda \in \Lambda} w(\lambda)\delta_\lambda$ which is almost-periodic in a sense explained in the following section. The Fourier transform $\hat{\mu}$ of μ (in the distributional sense) is a sum $\sum_0^\infty \nu_m$ where $\nu_m = \sum_{y \in \Lambda_m^*} w_m^*(y)\delta_y$ and $\Lambda_m^* = \Lambda^* + \cdots + \Lambda^*$ (m times). But the dual weights w_m^* have a rapid decay as m tends to infinity and only finitely many layers are observed in the diffraction pattern of a quasicrystal.

All these results can be found in my two books [**1**] and [**2**] which were published in 1970 and 1972.

2. Almost-periodic functions, Poisson summation formula and algebraic numbers.

Once for all, the Fourier transform of a function $f \in L^1(\mathbb{R}^n)$ will be defined by

$$\hat{f}(\xi) = \int e^{-ix \cdot \xi} f(x) \, dx \tag{2.1}$$

and the Fourier inversion formula reads

$$f(x) = (2\pi)^{-n} \int e^{ix \cdot \xi} \hat{f}(\xi) \, d\xi \, . \tag{2.2}$$

A continuous function $f : \mathbb{R}^n \to \mathbb{C}$ is almost-periodic (in the sense given by Bohr) if for each positive ε there exists a *relatively dense* set Λ_ε of *ε-almost periods* τ for f. These two concepts (relatively dense and ε-almost period) are now defined.

A subset $\Lambda \subset \mathbb{R}^n$ is *relatively dense* if there exists a positive R such that each ball with radius R (whatever be its center) contains at least a point λ in Λ.

An *ε-almost period* τ for f is defined by

$$\sup_{x\in\mathbb{R}^n} |f(x+\tau) - f(x)| \leq \varepsilon .$$

A more constructive approach to almost periodic functions is the following. First let us observe that any finite trigonometric sum $P(x) = \sum_{\lambda\in F} c(\lambda)e^{i\lambda\cdot x}$ is almost-periodic (F is a finite subset of $\mathbb{R}^n$). Next if f_j, $j = 0, 1, 2, \ldots$ is a sequence of almost-periodic functions and if $\|f - f_j\|_\infty \to 0$ ($j \to +\infty$) then f is also an almost-periodic function. Here and in what follows, $\|f\|_\infty = \sup_{x\in\mathbb{R}^n} |f(x)|$.

Finally a function $f : \mathbb{R}^n \to \mathbb{C}$ is almost-periodic if and only if, for each $\varepsilon > 0$, there exists a finite trigonometric sum $P_\varepsilon(x) = \sum_{\lambda\in F(\varepsilon)} c(\lambda, \varepsilon)e^{i\lambda\cdot x}$ such that $\|f - P_\varepsilon\|_\infty \leq \varepsilon$.

This approach will become a more constructive one if one knows how to construct $F(\varepsilon)$ and $P_\varepsilon(x)$ when f is a given almost-periodic function. For this purpose, we need to define the Fourier coefficients of an almost-periodic function.

Let $B(x, R)$ be the ball centered at x, with radius R and let $|B(x, R)| = c_n R^n$ be its volume. Then the limit

$$\mathcal{M}(f) = \lim_{R\to+\infty} |B(x, R)|^{-1} \int_{B(x,R)} f(y)\, dy \tag{2.3}$$

is attained uniformly in x for an almost periodic function f.

Moreover, for each $\omega \in \mathbb{R}^n$, $e^{-i\omega\cdot x} f(x)$ is also an almost-periodic function and the Fourier coefficients $\gamma(\omega)$ of f are defined as

$$\mathcal{M}(e^{-i\omega\cdot x} f(x)) = \gamma(\omega) . \tag{2.4}$$

If f is almost-periodic, so is $|f|^2$ and one has

$$\mathcal{M}(|f|^2) = \sum_{\omega\in S} |\gamma(\omega)|^2 . \tag{2.5}$$

This implies that the set S of those ω for which $\gamma(\omega) \neq 0$ is at most a numerable set. This set S is named the spectrum of f and one would expect that, in some sense,

$$f(x) = \sum_{\omega \in S} \gamma(\omega)\, e^{i\omega \cdot x} . \tag{2.6}$$

Summation procedures generalizing Cesaro summation have been fixed to give a meaning to (2.6). There exists a family $\beta_S(\varepsilon, \omega)$ of weight factors such that $0 \leq \beta_S(\varepsilon, \omega) \leq 1$, $\lim_{\varepsilon \downarrow 0} \beta_S(\varepsilon, \omega) = 1$ for each $\omega \in S$, $\beta_S(\varepsilon, \omega) = 0$ if ω does not belong to the finite set $F(\varepsilon)$ and finally $\|f - P_\varepsilon\|_\infty \to 0$ when ε tends to 0 and when

$$P_\varepsilon(x) = \sum_{\omega \in S} \beta_S(\varepsilon, \omega)\, \gamma(\omega)\, e^{i\omega \cdot x} . \tag{2.7}$$

As indicated by the subscript S, $\beta_S(\varepsilon, \omega)$ does not depend on f but only on the additive properties of S.

Let us return to the connection between quasicrystals and almost-periodic functions which has been alluded to in the introduction. If Λ is a quasicrystal, then for any positive ε there exists a second quasicrystal Λ_ε^* (named the ε-dual of Λ) such that

$$|e^{i\lambda \cdot y} - 1| \leq \varepsilon \qquad \text{for} \quad \lambda \in \Lambda \quad \text{and} \quad y \in \Lambda_\varepsilon^* . \tag{2.8}$$

This implies

$$\|f(x-y) - f(x)\|_\infty \leq C\varepsilon \|f\|_\infty \quad , \quad y \in \Lambda_\varepsilon^* \tag{2.9}$$

where f is *any* almost-periodic function whose spectrum lies in Λ and C is an absolute constant $\left(|f|_\infty = \sup_{x \in \mathbb{R}^n} |f(x)|\right)$. The key point is the fact that the ε-almost periods $y \in \Lambda_\varepsilon^*$ do not depend on f.

A second relation between quasicrystals and almost-periodic structures is provided by the notion of an almost periodic Radon measure.

Definition 3. *A (unbounded) Radon measure μ on $\mathbb{R}^n$ is almost periodic if the convolution product $\mu * g = f$ is an almost periodic function whenever g is a compactly supported continuous function.*

If μ is an almost-periodic measure, the closed graph theorem implies that $\sup_{x \in \mathbb{R}^n} \int_{B(x,R)} |d\mu| = C(R) < \infty$ for any $R > 0$. Therefore $\mu * g$ is an almost-periodic function for any g in the Schwartz class $\mathcal{S}(\mathbb{R}^n)$.

In a sense to be made precise, the theory of quasicrystals amounts to studying those almost-periodic measures μ which have the property that their (distributional) Fourier transform $\hat{\mu}$ is also an almost-periodic Radon measure.

Roughly speaking μ will be a (weighted) sum of Dirac masses δ_λ where λ runs over a quasicrystal Λ and $\hat{\mu}$ will be the corresponding (weighted) sum of Dirac masses δ_y where y runs over the dual quasicrystal Λ^*.

A trivial example is given by the ordinary Poisson formula. In that case, Λ is a lattice. A lattice $\Gamma \subset \mathbb{R}^n$ is a discrete subgroup with compact quotient. In other words $\Gamma = A(\mathbf{Z}^n)$ where A is an $n \times n$ invertible matrix. We define $\mathrm{vol}\,(\Gamma)$ as the volume of any fundamental domain of Γ. Then $\mathrm{vol}\,(\Gamma) = |\det A|$.

The dual lattice $\Gamma^* \subset \mathbb{R}^n$ is defined by $e^{iy\cdot x} = 1$ for every $x \in \Gamma$ and every $y \in \Gamma^*$.

Then the *Poisson summation* formula reads

$$\mathrm{vol}\,(\Gamma) \sum_{\gamma\in\Gamma} f(\gamma) = \sum_{\gamma\in\Gamma^*} \hat{f}(\gamma) \tag{2.10}$$

whenever f belongs to the Schwartz class $\mathcal{S}(\mathbb{R}^n)$. In other words, the Fourier transform of $\mu = \sum_{\gamma\in\Gamma} \delta_\gamma$ is $\hat{\mu} = \mathrm{vol}\,(\Gamma^*) \sum_{\gamma\in\Gamma^*} \delta_\gamma$ which provides a trivial example of an almost-periodic measure μ with an almost-periodic Fourier transform.

This section will be ended with the definition of some remarkable algebraic integers, the so-called Pisot and Salem numbers.

Definition 4. *A Pisot number is a real number $\theta > 1$ with the following two properties*

(2.11) θ *is an algebraic integer of degree* $n \geq 1$

(2.12) *the conjugates* $\theta_2, \ldots, \theta_n$ *of* θ *satisfy* $|\theta_2| < 1, \ldots, |\theta_n| < 1$.

For example, $2, 3, \ldots$ are Pisot numbers and the second condition is vacuous in that case. When the degree n of θ exceeds 1, we have $\theta^n + a_1\theta^{n-1} + \cdots + a_n = 0$ where $a_1 \in \mathbf{Z}, \ldots, a_n \in \mathbf{Z}$. Then $\theta_2, \ldots, \theta_n$ are the other solutions to this equation and can be either real or complex numbers.

Salem numbers are defined in the same way. One keeps condition (2.11) but replaces (2.12) by $|\theta_2| \leq 1, \ldots, |\theta_n| \leq 1$ with, at least, equality somewhere. Then n is even, $\theta_2 = \frac{1}{\theta}$ and $|\theta_3| = \cdots = |\theta_n| = 1$ ([4]).

3. Model sets and quasicrystals.

A *model set* $\Lambda \subset \mathbb{R}^n$ is defined the following way. One considers a lattice $D \subset \mathbb{R}^n \times \mathbb{R}^m$ where m is an integer and $m = 0$ is not excluded. If $(x, y) = X \in \mathbb{R}^n \times \mathbb{R}^m$, we write $x = p_1(X)$ and $y = p_2(X)$.

Let us assume that $p_1 : D \to p_1(D) \subset \mathbb{R}^n$ is a $1-1$ mapping and that $p_2(D)$ is a dense subgroup of $\mathbb{R}^m$. The latter condition is always satisfied when $m = 0$.

Definition 5. *Keeping these notations, let B be any bounded set in $\mathbb{R}^m$. Then the model set Λ defined by D and B is the collection of all $\lambda = p_1(d)$ such that $d \in D$ and $p_2(d) \in B$.*

The reader is referred to definition 4, page 48 of [**2**].

When $m = 0$, then $D = \Lambda$ is a lattice in $\mathbb{R}^n$. If $m \geq 1$ and if $p_2 : D \to p_2(D)$ is not a $1-1$ mapping, one easily shows that in a suitable coordinate system $\Lambda = \Lambda_1 \times \Lambda_2$ where $\Lambda_1 \subset \mathbb{R}^{n_1}$ is a lattice and $\Lambda_2 \subset \mathbb{R}^{n_2}$ is a model set. We have $n = n_1 + n_2$, $n_2 < n$ and the mapping p_2 associated to Λ_2 is $1-1$. It means that the general definition of model sets can be reduced to the special case when $p_2 : D \to p_2(D)$ is a $1-1$ mapping. In that special case, any subset Λ' of a model set Λ is still a model set.

Let us remind the reader of the definition of a Delaunay set Λ. Each ball of radius R_1 whatever be its location should contain at most one point in Λ and each ball of radius R_2 whatever be its location should contain at least one point in Λ ($R_2 > R_1 > 0$).

Definition 6. *A quasicrystal is a subset Λ of $\mathbb{R}^n$ fulfilling the following two conditions*

(3.1) Λ *is a Delaunay set*

(3.2) *there exists a finite set* F *contained in* $\mathbb{R}^n$ *such that* $\Lambda - \Lambda \subset \Lambda + F$.

Before going further let us make a few remarks on that definition. From (3.1) and (3.2) one immediately deduces that $\Lambda + F$ is still a Delaunay set. So is $\Lambda - \Lambda$ and moving one step further one obtains that $\Lambda - \Lambda - \Lambda$ is also a Delaunay set.

If Λ is a Delaunay set and if $\Lambda - \Lambda - \Lambda$ is also a Delaunay set, then one has $\Lambda - \Lambda \subset \Lambda + F$. The proof is obvious. Let $\lambda \in \Lambda$, $\lambda' \in \Lambda$; then $\lambda - \lambda' = \lambda'' + r$ where $\lambda'' \in \Lambda$ and $|r| \leq R_2$ (since Λ is a Delaunay set). Therefore $r = \lambda - \lambda' - \lambda'' \in \Lambda - \Lambda - \Lambda$ and $|r| \leq R_2$. Finally $r \in F$ which is a finite set since $\Lambda - \Lambda - \Lambda$ is a Delaunay set.

We can conclude

Lemma 1. *A quasicrystal is a Delaunay set Λ such that $\Lambda-\Lambda-\Lambda$ is also a Delaunay set.*

Property (3.1) in the definition of a quasicrystal implies that for each $R>0$ and each $\lambda_0 \in \Lambda$, the set $F(\lambda_0, R) = \{\lambda-\lambda_0 \ ; \ \lambda \in \Lambda \text{ and } |\lambda-\lambda_0| \leq R\}$ is finite. But property (3.2) tells us that there are only finitely many such finite sets $F(\lambda_0, R)$ when λ_0 runs over Λ, R being kept fixed. This is exactly a Penrose tiling property.

The following theorem relates model sets and quasicrystals.

Theorem 1. *Let $\Lambda \subset \mathbb{R}^n$ be a model set such that the corresponding bounded set B has a non empty interior. Then Λ is a quasicrystal.*

Conversely if Λ is a quasicrystal, there exists a finite set F and a model set Λ_0 such that $\Lambda \subset \Lambda_0 + F$. Moreover this model set Λ_0 corresponds to a bounded set B_0 with a non empty interior.

This theorem is better understood if the following definition is used.

Definition 7. *Two subsets Λ and Λ' of $\mathbb{R}^n$ are equivalent modulo finite sets if there exist two finite sets F and F' such that $\Lambda \subset \Lambda'+F'$ and $\Lambda' \subset \Lambda+F$. One then writes $\Lambda \sim \Lambda'$.*

If one keeps p_1, p_2 and D fixed and lets B vary, all the model sets Λ_B associated to bounded sets B with non empty interior will be equivalent. It is also true that any set which is equivalent to a quasicrystal will still be a quasicrystal. This remark is no longer true for model sets. A counterexample is given by $\mathbf{Z} + \{0, \sqrt{2}\}$ which is not a model set while $\mathbf{Z}$ is a model set. That is the reason why theorem 1 does not say that any quasicrystal is a model set.

If two equivalent model sets are identified, one is reduced to the case where B is the unit ball in definition 5.

4. Quasicrystals and Diophantine approximation.

In this section Λ will denote a model set for which the corresponding bounded set B is Riemann-integrable. It means that the Lebesgue measure of $\overline{B}$ (the closure of B) equals the Lebesgue measure of $\operatorname{int}(B)$ (the interior of B).

Let $D^* \subset \mathbb{R}^n \times \mathbb{R}^m$ be the dual lattice of D. It means that D^* is the collection of all $(\xi, \eta) \in \mathbb{R}^n \times \mathbb{R}^m$ such that $x \cdot \xi + y \cdot \eta \in 2\pi\mathbf{Z}$ for each

$(x, y) \in D$. Then $p_1 : D^* \to p_1(D^*)$ is still a $1-1$ mapping while $p_2(D^*)$ is still dense in $\mathbb{R}^m$.

A first observation is an asymptotic form of the Delaunay property. If Λ is a model set for which B is Riemann-integrable, then each ball $B(x, R)$ centered at x with radius R contains exactly $cR^n + o(R^n)$ points in Λ where $c = \frac{c_n}{\text{vol } D}$ and where the little o is uniform in x.

If c_λ, $\lambda \in \Lambda$, is any bounded sequence, its mean value over Λ (if it exists) will be defined as the uniform limit as R tends to infinity of

$$\frac{\sum_{\lambda \in B(x,R)} c_\lambda}{\sharp \{B(x, R) \cap \Lambda\}} . \tag{4.1}$$

With these definitions in mind, one can state the following theorem ([**2**], page 159).

Theorem 2. *Let Λ be a model set for which B is Riemann-integrable with a non empty interior. Let $\omega \in \mathbb{R}^n$ and let us assume that $\omega \notin p_1(D^*)$. Then the mean value over Λ of $e^{i\omega\cdot\lambda}$ is 0.*

If $\omega = p_1(d^*)$ and $d^* \in D^*$, then the mean value of $e^{i\omega\cdot\lambda}$ over Λ is given by $|B|^{-1}\hat{\mathbf{1}}_B(p_2(d^*))$ where $\mathbf{1}_B$ is the indicator function of B, $\hat{\mathbf{1}}_B$ denotes the Fourier transform of this indicator function and $|B|$ is the Lebesgue measure of B.

Theorem 2 paves the road to the definition of the dual lattice. Let us assume that $|e^{i\omega\cdot\lambda} - 1| \leq 1$ uniformly over Λ. Then $Re(e^{i\omega\cdot\lambda}) \geq 1$ and by convexity the same inequality will be valid for any average of $e^{i\omega\cdot\lambda}$, $\lambda \in \Lambda$. Therefore $\omega \in p_1(D^*)$ (otherwise the mean value of $e^{i\omega\cdot\lambda}$ would be 0).

If $\omega = p_1(d^*)$ while $\lambda = p_1(d)$, then

$$\omega \cdot \lambda + p_2(d) \cdot p_2(d^*) \in 2\pi\mathbf{Z} \tag{4.2}$$

and

$$\exp(i\omega \cdot \lambda) = \exp(-ip_2(d) \cdot p_2(d^*)) . \tag{4.3}$$

Since $p_2(D)$ is dense in $\mathbb{R}^m$, so is $B \cap p_2(D)$ in B and $|\exp(ip_2(d)\cdot p_2(d^*)) - 1| \leq 1$ for $p_2(d) \in B$ implies $|\exp(iy \cdot p_2(d^*) - 1| \leq 1$ for any $y \in B$.

Since B has a non empty interior, that forces $p_2(d^*)$ to belong to a bounded set B^*.

Therefore $\omega = p_1(d^*)$ belongs to a model set $\Lambda^*(B^*)$. Let us observe that $|e^{i\omega\cdot\lambda} - 1| \leq \rho < \sqrt{2}$ uniformly over Λ would also lead to $\omega \in \Lambda^*(B^*)$ but this argument breaks down if $\rho > \sqrt{2}$.

We do not know the optimal value of ρ ensuring that $|e^{i\omega\cdot\lambda}-1| \leq \rho$ uniformly over a given model set Λ implies $\omega \in \Lambda^*$ where Λ^* is also a model set. Let us assume that B is a closed convex set, that B is Riemann integrable and $0 \in B$.

Then $|e^{i\omega\cdot\lambda}-1| \leq 1$ uniformly over Λ is equivalent to $\omega = p_1(d^*)$, $d^* \in D^*$ and $|y \cdot p_2(d^*)| \leq \pi/3$ for each $y \in B$.

Let B^* be the dual convex set defined by $z \in B^*$ iff $\sup_{y\in B} |z \cdot y| \leq \pi/3$. Then $p_2(d^*) \in B^*$.

A full symmetry between a model set Λ and its dual model set Λ^* is then given by the following theorem.

Theorem 3. *If B is Riemann integrable, closed, convex and symmetric around 0, if Λ is the model set defined by $\{\lambda = p_1(d)\ ;\ d \in D$ and $p_2(d) \in B\}$, then the dual $\Lambda^* = \{y \in \mathbb{R}^n\ ;\ |e^{i\lambda\cdot y}-1| \leq 1\ ,\ \lambda \in \Lambda\}$ is also a model set given by*

$$\Lambda^* = \left\{\lambda = p_1(d^*)\ ;\ d^* \in D^* \text{ and } p_2(d^*) \in B^*\right\}$$

and $(\Lambda^*)^* = \Lambda$.

Let us now compute the ε-dual set Λ_ε^* defined by $|e^{iy\cdot\lambda}-1| \leq \varepsilon$, $\lambda \in \Lambda$, when $\varepsilon \in (0,1)$. We immediately obtain

$$y \in \Lambda_\varepsilon^* \Leftrightarrow y = p_1(d^*) \quad , \quad d^* \in D^* \quad \text{and} \quad p_2(d^*) \in \eta B^* \tag{4.4}$$

where $0 < \eta < 1$ and $\sin\left(\eta\,\frac{\pi}{6}\right) = \frac{\varepsilon}{2}$.

Therefore Λ_ε^* is also a model set.

This observation paves the way to the following definition.

Definition 8. *A subset Λ of $\mathbb{R}^n$ is harmonious if for each $\varepsilon > 0$, the ε-dual set Λ_ε^* is relatively dense in $\mathbb{R}^n$.*

Any finite set F is harmonious. If Λ is harmonious and if F is finite, then $\Lambda+F$ is still harmonious. If Λ is harmonious, so is $\Lambda-\Lambda$. Finally any harmonious set is uniformly discrete : there exists a positive δ such that $\lambda \in \Lambda$, $\lambda' \in \Lambda$ and $\lambda' \neq \lambda$ imply $|\lambda'-\lambda| \geq \delta$.

Any model set is harmonious and theorem 1 implies that any quasicrystal is harmonious.

Let us prove the following observation.

Theorem 4. *Let Λ be a Delaunay set in $\mathbb{R}^n$. Then Λ is a quasicrystal if and only if Λ is harmonious.*

If Λ is harmonious and is a Delaunay set, then $\Lambda_3 = \Lambda - \Lambda - \Lambda$ is also harmonious. Therefore Λ_3 is uniformly discrete and is a Delaunay set. Lemma 1 shows that Λ is a quasicrystal.

Let us return to the relationship between the arithmetical properties of a set Λ and the uniform almost periodicity of the linear space C_Λ of all trigonometric sums $f(x) = \sum_{\lambda\in\Lambda} c_\lambda\, e^{i\lambda\cdot x}$ whose frequencies belong to Λ.

I wanted to know whether for each positive ε there exists a Delaunay set T_ε such that

$$\tau \in T_\varepsilon \;\Rightarrow\; \|f(x-\tau)-f(x)\|_\infty \le \varepsilon\|f\|_\infty \quad , \quad f \in C_\Lambda\,. \tag{4.5}$$

If one picks $f(x) = e^{i\lambda\cdot x}$, then one obtains $|e^{i\lambda\cdot\tau}-1| \le \varepsilon$ for each $\tau \in T_\varepsilon$. Therefore Λ is harmonious. Conversely if Λ is harmonious, a clever lemma due to N. Varopoulos implies that $\|f(x-\tau)-f(x)\|_\infty \le C\varepsilon\|f\|_\infty$ where C is an absolute constant and $f \in C_\Lambda$.

The relationship between Pisot numbers, Salem numbers and harmonious sets is given by the two following theorems.

Theorem 5. *Let $\theta > 1$ be a real number. Then the set $S = \{1, \theta, \theta^2, \ldots, \theta^j, \ldots\}$ is harmonious if and only if θ is a Pisot or is a Salem number.*

The proof is given in [**2**], p. 37.

A simple corollary is the following observation.

Theorem 6. *If Λ is a quasicrystal, if $\theta > 1$ is a real number and if $\theta\Lambda \subset \Lambda$, then θ is either a Pisot number or is a Salem number.*

Conversely for each dimension n and each Pisot or Salem number θ, there exists a quasicrystal Λ in $\mathbb{R}^n$ such that $\theta\Lambda \subset \Lambda$.

Let us prove theorem 6.

Let $\lambda_0 \in \Lambda$, $\lambda_0 \neq 0$. Then $\theta\Lambda \subset \Lambda$ implies that $\lambda_0 S$ is contained in Λ when S is defined as in theorem 5. But a subset of a harmonious set is obviously a harmonious set. Therefore $\lambda_0 S$ is a harmonious set in $\mathbb{R}^n$ and also in the line $\lambda_0\mathbb{R}$. Finally S is harmonious and theorem 5 can be used.

. Conversely let θ be either a Pisot or a Salem number and let $K = Q(\theta)$ be the algebraic number field generated by θ. Let $\theta_2, \ldots, \theta_n$ be the conjugates of θ and $\sigma_2 : K \to \mathbb{C}, \ldots, \sigma_n : K \to \mathbb{C}$ be the $\mathbb{Q}$-isomorphisms defined by $\sigma_2(\theta) = \theta_2, \ldots, \sigma_n(\theta) = \theta_n$).

The ring Ω of all algebraic integers contained in K admits a direct sum decomposition given by

$$\Omega \;=\; \mathbf{Z}\omega_1 + \cdots + \mathbf{Z}\omega_n\,. \tag{4.6}$$

Let Λ be the collection of all algebraic integers $\lambda \in \Omega$ such that $|\sigma_2(\lambda)| \le 1, \ldots, |\sigma_n(\lambda)| \le 1$. Then Λ is a model set for which $m = n-1$, $D = \Omega$, $p_2 = (\sigma_2, \ldots, \sigma_n)$ and p_1 is the identity mapping.

Moreover Λ is closed under multiplication : if $\lambda \in \Lambda$ and $\lambda' \in \Lambda$, then $\lambda\lambda' \in \Lambda$. Therefore $\theta\Lambda \subset \Lambda$.

We have obtained a one dimensional model set Λ such that $\theta\Lambda \subset \Lambda$. If one needs an n-dimensional model set, it suffices to take $\Lambda \times \cdots \times \Lambda$ (n times).

5. Poisson summation formula and quasicrystals.

Let Λ be a model set defined as above by $D \subset \mathbb{R}^n \times \mathbb{R}^m$ and $B \subset \mathbb{R}^m$. Let us assume this bounded set B Riemann integrable and let φ denote any $C_0^\infty(\mathbb{R}^m)$ function vanishing outside B.

Using this function φ, weights $w(\lambda)$ are defined on the model set Λ by $w(p_1(d)) = \varphi(p_2(d))$, $d \in D$. If φ were the indicator function of B (which is not smooth), we would have $w(\lambda) = 1$ on Λ and $w(p_1(d)) = 0$ if $p_1(d) \notin \Lambda$.

With these notations, one obtains

Theorem 7. *Let μ be the sum $\sum_{\lambda\in\Lambda} w(\lambda)\delta_\lambda$ of Dirac masses over Λ where the weighting factors are defined as above. Then the Fourier transform in the distributional sense $\hat{\mu}$ of μ is given by*

$$\hat{\mu} = \sum_{D^*} \omega(p_2(d^*))\, \delta_{p_1(d^*)} \tag{5.1}$$

where the dual weights $\omega(p_2(d^*))$ are defined by

$$\omega(y) = \frac{(2\pi)^n}{\operatorname{vol} D}\, \hat{\varphi}(-y) \quad , \quad y = p_2(d^*)\ ,\ d^* \in D^*\ . \tag{5.2}$$

If one could find a testing function $\varphi \in C_0^\infty(\mathbb{R}^n)$ with a compactly supported Fourier transform, $\hat{\mu}$ would then be supported by the dual quasicrystal Λ^*. This is not the case. But let us consider the case where B is a convex set, symmetric around 0 and Riemann-integrable. Then the dual quasicrystal Λ^* is defined by the dual convex set B^*. For each integer $m \geq 1$, let us define Λ_m^* with mB^* instead of B^*, p_1, p_2 and D^* being kept fixed.

Then theorem 7 yields $\hat{\mu} = \sum_{m\geq 1} \nu_m$ where ν_m is supported by Λ_m^* and where this series has a rapid decay at infinity. More precisely

$$|\nu_m|(B) = 0(m^{-N}) \tag{5.3}$$

for every ball B and each integer N.

This rapid decay explains why only finitely many layers Λ_m^* are observed in the diffraction pattern of a quasicrystal.

6. Conclusion.

To my opinion, the most sensible definition of a quasicrystal is given in terms of Diophantine approximation : a quasicrystal Λ is a Delaunay set such that the ε-dual sets Λ_ε^* are also Delaunay sets whenever $0 < \varepsilon \leq 1$. The ε-dual set Λ_ε^* is defined by $|e^{iy\cdot\lambda}-1| \leq \epsilon$, $\lambda \in \Lambda$, $y \in \Lambda_\varepsilon^*$.

This approach is appealing since it gives an interesting heuristic to the diffraction pattern. Indeed if the weights $w(\lambda)$, $\lambda \in \Lambda$, are nonnegative, the corresponding sum $\sum_{\lambda\in\Lambda} w(\lambda)e^{i\lambda\cdot y}$ attains its largest value on Λ^* and is everywhere larger than a half of this largest value on the dual quasicrystal Λ^*.

This explains why the Fourier transform of the measure μ in theorem 6 peaks on the dual quasicrystal Λ^*.

The Diophantine approximation definition of quasicrystals also explains the role played by Pisot and Salem numbers as inflation factors of quasicrystals. This property is used in a systematic way in J. Patera's work where the Pisot number $\tau = \frac{1+\sqrt{5}}{2}$ plays a key role.

Finally this Diophantine approximation characterization is exactly what is needed for proving that almost-periodic functions with frequencies in a given quasicrystal Λ behave like periodic functions with frequencies in a given lattice.

The second definition of a quasicrystal is the celebrated "cut and projection" algorithm. I was led to this definition when I tried to generalize examples of the type $\Lambda_m = \left\{\sum_{k\geq 0}\varepsilon_k\theta^k \ ; \ \varepsilon_k \in \mathbf{Z} \ , \ |\varepsilon_k| \leq m\right\}$ where $m \geq 1$ is a given integer, θ is a Pisot number and $\varepsilon_k = 0$ when k is large enough. In such an example each $\lambda \in \Lambda$ is an algebraic integer and its conjugates $\lambda_2, \ldots, \lambda_n$ will satisfy

$$|\lambda_2| \leq \frac{m}{1-|\theta_2|} \quad , \ldots , \quad |\lambda_m| \leq \frac{m}{1-|\theta_n|} \, .$$

The third definition of a quasicrystal is given by $\Lambda-\Lambda \subset \Lambda+F$ (F finite) *and* Λ *is a Delaunay set.* This approach to quasicrystals is striking by its elegant simplicity. But this approach is deceiving since it does not directly yield the Diophantine approximation characterization of quasicrystals. One has to go through the "cut and projection" method (theorem 1).

One is tempted to relate this additive characterization of quasicrystals to the definition of pentacrystals which was introduced by J. Patera. Patera defines a pentacrystal by the following two conditions : Λ is a Delaunay set

and $\tau^2\Lambda - \tau\Lambda = \Lambda$ where $\tau = \frac{1+\sqrt{5}}{2}$. Pentacrystals are examples of quasicrystals and τ is an example of a Pisot number.

Admiring the spectacular examples of quasicrystals which were designed by J. Patera, I better understand why my early work was never given the credit it deserved. A systematic search for all quasicrystals was needed for solving the problem I was attacking. The specific quasicrystals designed by J. Patera would have been useless in my own work. They are based on a specific Pisot number while I needed any Pisot number.

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